

“MANY **SPECIES** OF ANIMALS HAVE BEEN **LOST** OUT OF THE WORLD, WHICH PHILOSOPHERS AND DIVINES ARE **UNWILLING TO ADMIT...**”

John Ray, English naturalist, from *Three physico-theological discourses, concerning the primitive Chaos, and creation of the world*, 1713

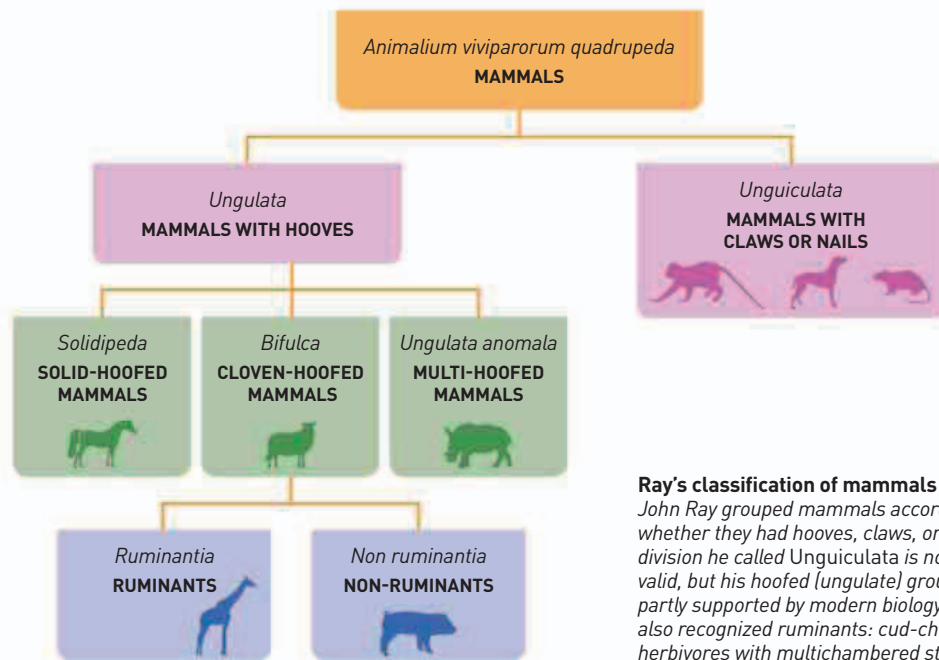
the **first scientific classification of animals**. He identified mammals as viviparous (giving birth to live young) quadrupeds, and placed them into groups according to structures such as feet and teeth.

In the same year, Belgian physician **Philip Verheyen** (1648–1711) published his illustrated *Anatomy of the Human Body*, which would become a standard textbook on the subject in European universities. In this work, Verheyen introduced the term Achilles tendon for the structure at the back of the leg,

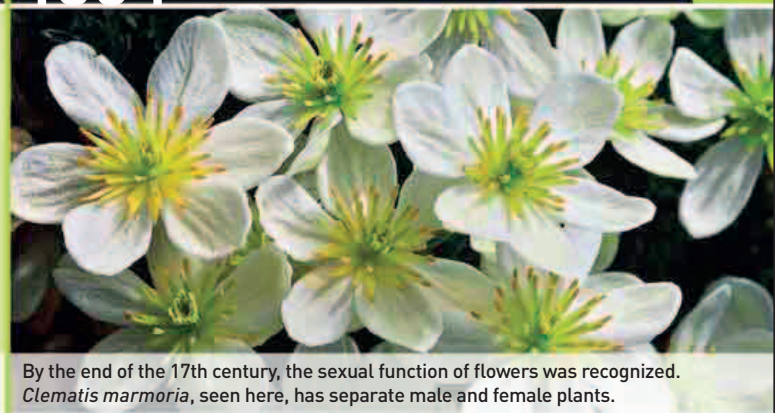
named for the Greek hero killed by an arrow wound in his heel. Opportunistically, Verheyen had been able to dissect his own left leg, which had been amputated because of illness nearly 20 years before. Verheyen had insisted on preserving the limb so he could study it. Based on personal experience, he was one of the first physicians to report **phantom limb phenomenon**: the sensation that an amputated limb is still attached to the body.

Also in 1693, English astronomer and mathematician **Edmond Halley** was the first

person to draw up **life annuity charts** based on mortality tables. Thirty years earlier, a shopkeeper called John Graunt had produced “life-tables” as part of a scheme to monitor the advance of bubonic plague, but Halley had the mathematical skills to carry out a more sophisticated analysis. Using data on births and deaths from the European city of Breslau, he estimated the city’s population size and the probabilities of its citizens surviving to particular ages. The study became a model for future demographic investigations.



Ray's classification of mammals
John Ray grouped mammals according to whether they had hooves, claws, or nails. The division he called Unguiculata is no longer valid, but his hoofed (ungulate) groups are partly supported by modern biology. He also recognized ruminants: cud-chewing herbivores with multichambered stomachs.



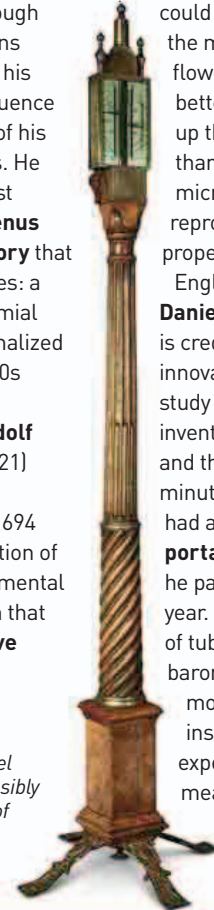
By the end of the 17th century, the sexual function of flowers was recognized. *Clematis marmoria*, seen here, has separate male and female plants.

BY THE END OF THE 17TH CENTURY THE WORK OF SEVERAL NATURALISTS had revealed some of the secrets of flowering plants. In 1694, French botanist **Joseph Pitton de Tournefort** (1656–1708) published a classification system that was based on the structure of flowers and fruits, as well as leaves and roots. Although Tournefort’s conclusions were often misguided, his work had a lasting influence because of the clarity of his species-level accounts. He was also one of the first botanists to use the **genus as a taxonomic category** that included similar species: a forerunner of the binomial system of naming formalized by Linnaeus in the 1700s (see 1733–39).

German botanist **Rudolf Camerarius** (1665–1721) went much further in studying flowers. His 1694 paper on the reproduction of plants provided experimental evidence for the notion that not only did **plants have**

sexual organs, but that pollen was the agent of male fertilization. Camerarius observed that female shoots separated from male shoots often failed to set seed, and that when pollen-producing stamens were removed, no seeds were produced at all. But he was frustrated by the fact that he could not probe deeper into the minute functions of flowers. It would not be until better microscopes opened up the world of cells more than a century later that the microscopic basis of plant reproduction could be properly explored.

English instrument maker **Daniel Quare** (1649–1724) is credited with a number of innovations in horology (the study of time), including the invention of repeating watches and the introduction of the minute hand. By 1694, he had also produced the **first portable barometer**, which he patented the following year. Until then, the system of tubes associated with a barometer was not easily moved, but a portable instrument could allow experimentalists to measure atmospheric pressure in places such as mines or mountains.



Portable barometer
The construction of Daniel Quare’s barometer ostensibly allowed free movement of the instrument without letting air in or spilling its mercury.

1693 English naturalist John Ray makes the first scientific classification of animals

1693 Belgian physician Philip Verheyen publishes a book on human anatomy

1693 English astronomer Edmond Halley creates the first life annuity charts

French botanist Joseph Tournefort uses flower structure as the basis for classification

German botanist Rudolf Camerarius publishes a treatise on the sexuality of plants

English instrument-maker Daniel Quare produces a portable barometer